

NAZIRU HALILU

Curriculum Vitae — Academic Format

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RESEARCH INTERESTS

Computer vision and geospatial machine learning for digital agriculture; self-supervised and contrastive learning for agronomic imaging; radiometric harmonization and vegetation indexing from UAV and satellite imagery; geostatistics and precision agriculture mapping (regression kriging, singularity indices); hybrid physics-informed deep learning for hydrological and environmental process modelling; closed-loop sensing and control for smart irrigation and greenhouse automation.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture 2025 to expected 2027

Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece

Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours 2019 to 2024

Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort

National Diploma, Agricultural Engineering Technology, Distinction 2016 to 2018

Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort

PUBLICATIONS

Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *International Journal of Research and Scientific Innovation (IJRSI)*, 13(7), 2920–2950. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. African Conf. on Precision Agriculture (AfCPA)*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (A case study of Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

Submitted / Under Review

- **Halilu, N.,** Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL): A Theoretical Framework and Baseline Evaluation Protocol for Agronomic Imaging. *Biosystems Engineering* (Elsevier). Under review. [DOI](#) | [Code](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). *Environmental Monitoring and Assessment* (Springer Nature). Under review. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy* (Elsevier). Under review. [DOI](#) | [Code](#)

Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging for Spatio-Temporal Soil NPK Crop Stress Yield-Prediction. *Zenodo Preprint* (v1.0.1). [DOI](#) | [Code](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction for Radiometrically Harmonized Multi-Temporal Agricultural Vegetation Monitoring. *Zenodo*. [DOI](#) | [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*. [DOI](#)

RESEARCH EXPERIENCE

EMJMD Research Scholar — Public University of Navarre, Spain

Sep 2025 to Jan 2026

Term 1: Technological bases for agricultural sustainability

- Designed STADOS, a five-stage radiometric harmonization framework, cutting temporal NDVI variance by 39% and 31% on real Sentinel-2 imagery over Portuguese corn parcels.
- Built a 14-module open-source RGB/multispectral vegetation indexing and GLCM texture analysis pipeline, reaching 33.8% balanced accuracy against a 4.55% chance baseline.
- Built a Bayesian-grounded digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models for gully erosion biocontrol (hold-out R-squared = 0.90).

EMJMD Research Scholar — UTAD, Portugal

Feb 2026 to Jun 2026

Term 2: Precision agriculture for sustainable crop production

- Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty by roughly 20% across 17,864 soil samples.
- Integrated GIS, remote sensing, and ML pipelines across three cropping systems, achieving R-squared between 0.978 and 0.995 and a 2.69-tonne reduction in lime demand.

Summer Intern, Computer Vision Group — Ahmadu Bello University

Jun 2026 to Aug 2026

Dept. of Computer Engineering

- Extended the GACL framework with four new geometry-invariant components, reaching 64.4% accuracy (macro AUC 0.882) across 19 disease classes.

AWARDS AND HONORS

- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural & Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- PTDF Undergraduate Fellow (2023)

TEACHING AND MENTORSHIP

- Undergraduate Course Tutor, ABU Zaria (NAMET), 2020–2024: Differential Equations & Transforms, Irrigation System Design, Computer Applications in Agric. Engineering.
- Deputy Coordinator, KOJYAF, 2020–2025: Mentorship programs preparing secondary students for JAMB entrance examinations.
- Virtual Secondary School Teacher, 2024–present: Remote instruction in Physics and Mathematics.

PROFESSIONAL MEMBERSHIPS

American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

REFERENCES

Prof. Javier Casalí Sarasibar — Department of Agricultural Engineering and Bioscience, Public University of Navarre, Spain · jcs@unavarra.es

Prof. Konstantinos Soulis — Agricultural University of Athens, Greece · soco@aua.gr

Dr. Ibrahim Babangida Dalha — Ahmadu Bello University, Nigeria · ibdalha@abu.edu.ng